Diamond Price Prediction: Midterm Report

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1 Summary of Findings

The project models diamond price as a supervised regression problem using the standard diamonds dataset. At the midterm stage, the strongest model is XGBoost: it reaches a 5-fold cross-validated RMSE of USD 526.9 ± 10.3 and MAE of USD 262.1 on the training split. Random Forest is close, with RMSE USD 552.5 ± 9.6, while Linear Regression is much weaker at RMSE USD 1,467.4 ± 153.5. All trained models substantially improve on the mean-price baseline.

The main modeling finding is that nonlinear tree ensembles fit this dataset far better than a linear model. The gain from Random Forest to XGBoost is smaller than the gain from Linear Regression to Random Forest, so XGBoost is the current leader but Random Forest remains a credible alternative. The test set closely matches their cross-validation scores, indicating excellent generalization and no strong evidence of overfitting.

2 Dataset Findings

The raw dataset contains 53,940 rows and 10 columns: `price`, `carat`, `cut`, `color`, `clarity`, `depth`, `table`, `x`, `y`, and `z`. The target is strictly positive, ranging from USD 326 to USD 18,823. No missing values were found in the required columns.

Cleaning removed 168 rows, or approximately 0.31% of the raw dataset. The removed records consisted of 20 rows with zero physical dimensions, 3 extreme dimension outliers, and 145 duplicate rows after invalid dimensional records were excluded. This is a small removal rate, so the cleaning step improves data validity without materially reducing the size of the training sample. The final cleaned dataset contains 53,772 rows.

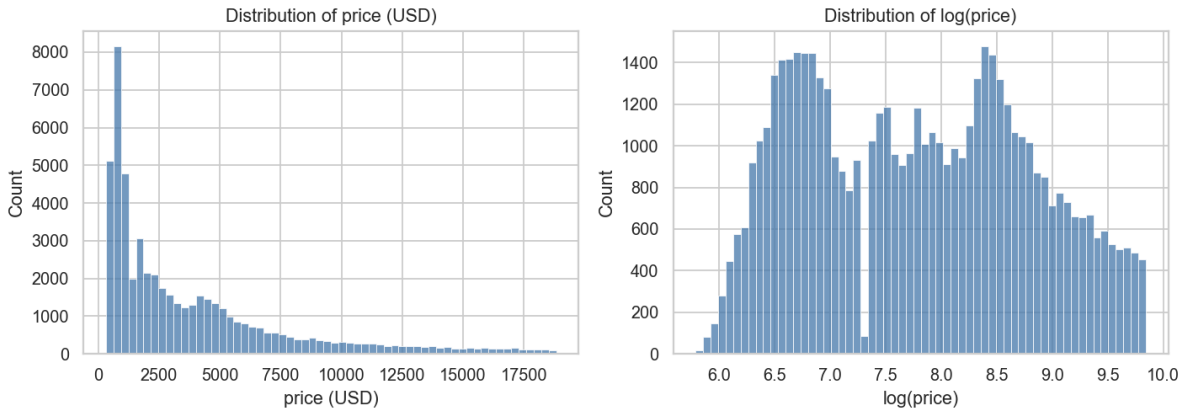


Figure 1: Distribution of price and log(price).

The target distribution is strongly right-skewed. Most diamonds are comparatively low-priced, while a smaller number of high-value diamonds extend the upper tail. This supports the use of a log-transformed target during model fitting: it reduces the dominance of expensive diamonds during training while still allowing final errors to be reported in USD after inverse transformation.



Figure 2: Carat versus price, sampled for readability and coloured by clarity.

Carat has the clearest visible relationship with price. The relationship is nonlinear: prices rise slowly for small stones, then increase more sharply for larger stones. The spread also increases as carat increases, indicating that quality grades and proportions matter more at higher price levels. This pattern is consistent with the later model results: a purely linear model captures the general trend, but tree ensembles capture nonlinear interactions more effectively.

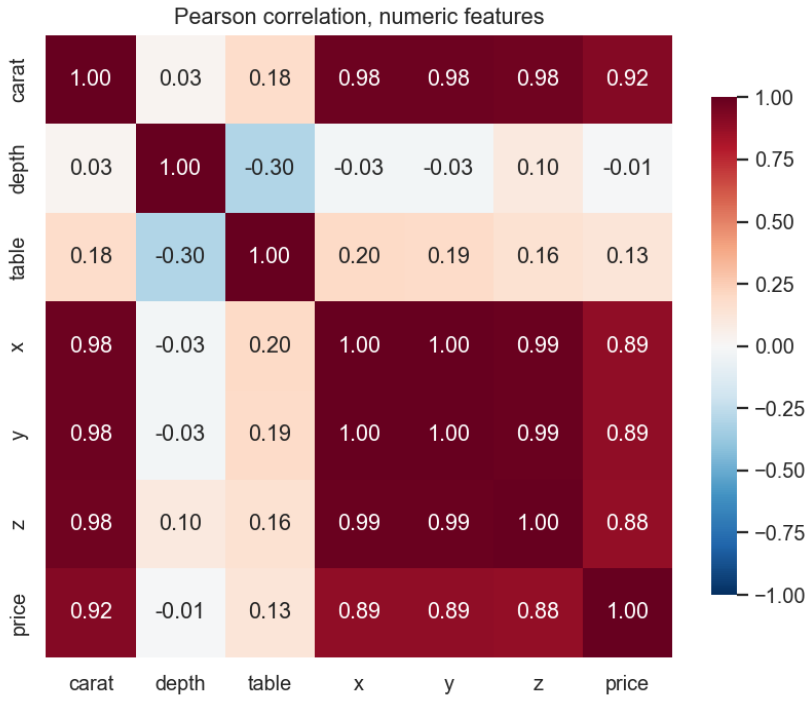


Figure 3: Correlation heatmap for numeric features.

The correlation matrix shows that `carat`, `x`, `y`, and `z` are highly collinear. They all describe stone size from different angles. This is useful for prediction but complicates interpretation, because a model can assign importance to one size proxy even when the underlying signal is shared across several correlated variables.

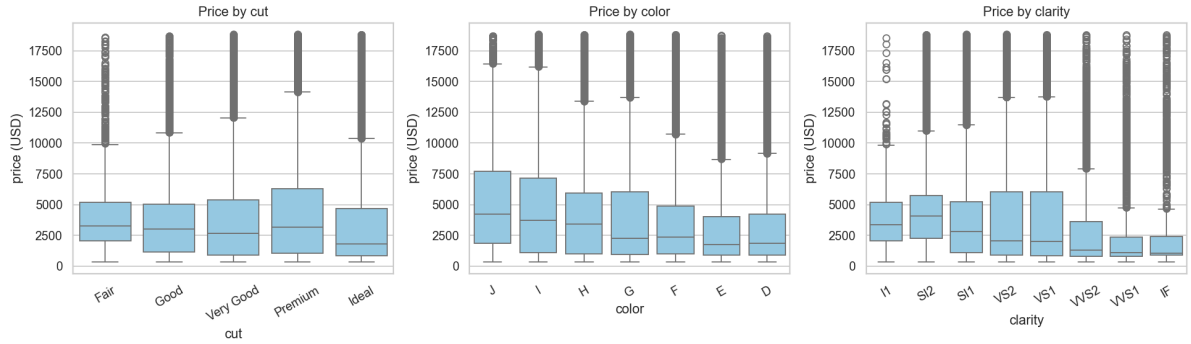


Figure 4: Price distributions by cut, color, and clarity.

The categorical quality grades show visible differences in price distributions, but their raw boxplots are affected by the distribution of stone size inside each category. For example, a high-quality category does not automatically have the highest raw median price if it contains many smaller stones. These plots therefore support including the categorical variables, but they do not by themselves establish a causal ranking of price effects.

3 Modeling Approach

The cleaned dataset was split into 80% training data and 20% held-out test data using a fixed random seed and stratification on carat quintiles.

The preprocessing pipeline encodes `cut`, `color`, and `clarity` as ordinal variables using their domain order. The `carat` feature is transformed with `log1p`. Numeric features are standardized for Linear Regression and passed through unscaled for tree-based models. The target is log-transformed during fitting and inverse-transformed before metric computation, so all reported errors are on the original USD scale.

Four models are compared: a mean-price baseline, Linear Regression, Random Forest with 200 trees, and XGBoost with 400 trees, learning rate 0.3 and depth 6. Hyperparameter tuning is not part of the midterm result; the purpose here is to establish a reproducible baseline and compare the default model families fairly under the same cross-validation protocol.

4 Model Results

Table 1: Training-split 5-fold cross-validation results. Lower RMSE, MAE, and MAPE are better; higher R^2 is better. All metrics report the mean ± standard deviation across the 5 folds.

Model	RMSE (USD)	MAE (USD)	R^2	MAPE
Mean baseline	3971.6 ± 42.2	3019.9 ± 16.5	-0.0002 ± 0.0001	187.5% ± 3.3%
Linear Regression	1467.4 ± 153.5	559.5 ± 12.4	0.8618 ± 0.0292	12.7% ± 0.1%
Random Forest	552.6 ± 9.6	271.9 ± 3.6	0.9806 ± 0.0006	6.5% ± 0.1%
XGBoost	**526.9 ± 10.3**	**262.1 ± 3.4**	**0.9824 ± 0.0005**	**6.1% ± 0.1%**

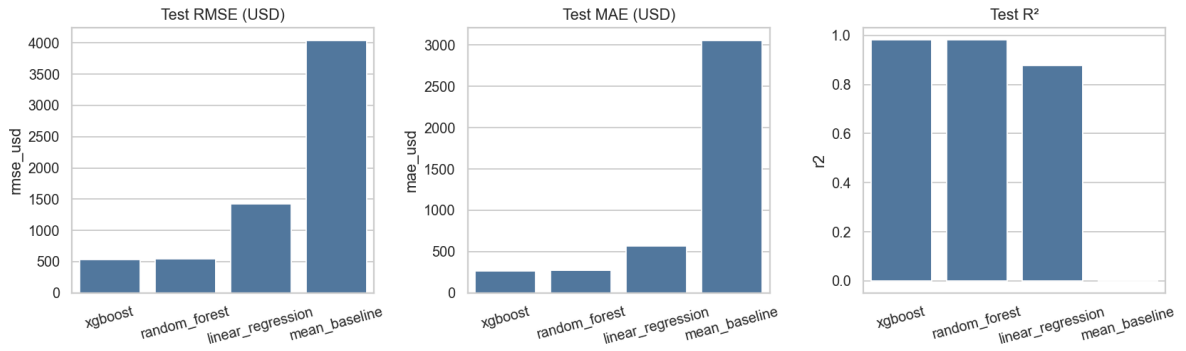


Figure 5: Cross-validation metric comparison.

The mean baseline performs poorly, with RMSE close to USD 4,000 and negative R^2, confirming that predicting the same average price for every diamond is not informative. Linear Regression reduces RMSE by about 63% compared with the baseline, which shows that the encoded features carry strong predictive signal. However, its remaining error is still high, especially compared with the tree ensembles.

Random Forest reduces RMSE by a further 62% compared with Linear Regression. This is the main performance jump in the current experiment and indicates that nonlinear splits and feature interactions are central to the task. XGBoost gives the best overall result, reducing RMSE by another 4.6% compared with Random Forest and achieving the best MAE, R^2, and

MAPE. The improvement over Random Forest is real in the current cross-validation table, but it is modest. The two tree models remain close enough that both are relevant for final tuning.

5 Learning Behavior and Error Analysis

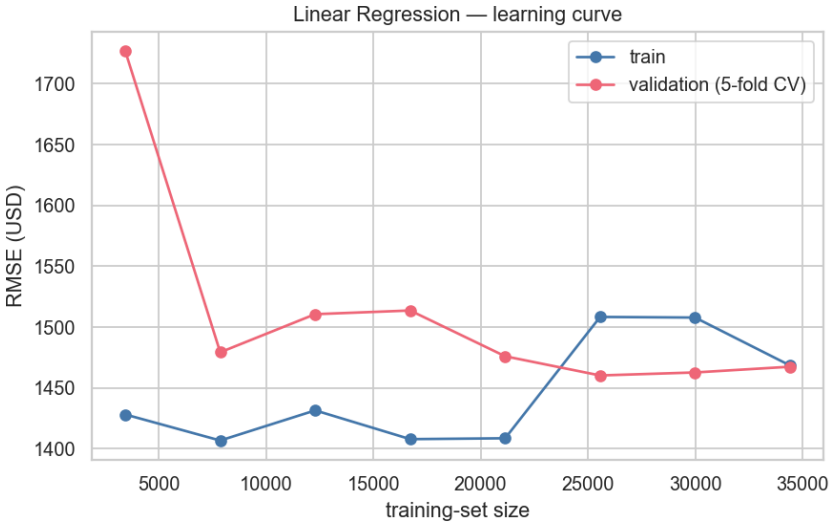


Figure 6: Linear Regression learning curve.

The Linear Regression learning curve stabilizes at a validation RMSE near USD 1,467. Adding more training data does not close the gap to the tree ensembles, which points to limited model capacity rather than insufficient sample size. This matches the EDA: the price relationship is nonlinear and depends on interactions between size, proportions, and quality grades.

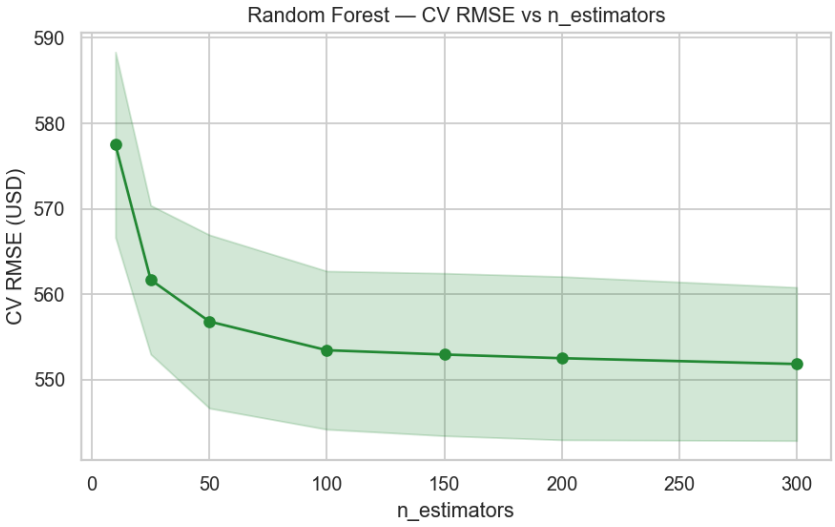


Figure 7: Random Forest cross-validated RMSE versus number of trees.

Random Forest improves quickly as the number of trees increases, then the curve flattens around the 100–200 tree range. The midterm setting of 200 trees is therefore a reasonable default: it captures most of the ensemble benefit without unnecessary training cost.

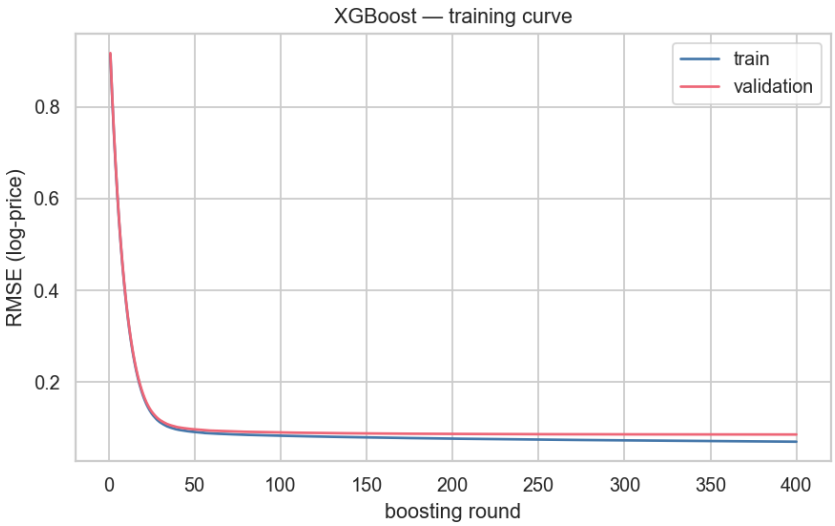


Figure 8: XGBoost training and validation RMSE over boosting rounds.

The XGBoost curve shows continuing reduction in training error and a weaker validation improvement after the earlier boosting rounds. The number of boosting rounds and the regularization settings are therefore important tuning parameters for the final phase. The current model is already strong, but the curve indicates room to make it more controlled.

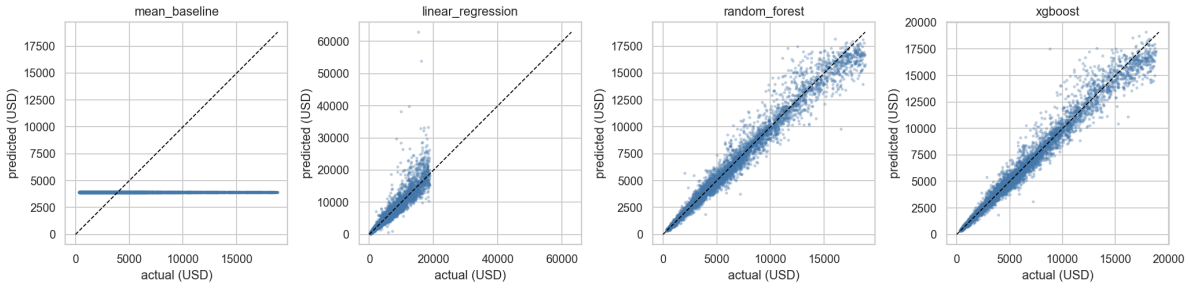


Figure 9: Cross-validated predicted versus actual prices.

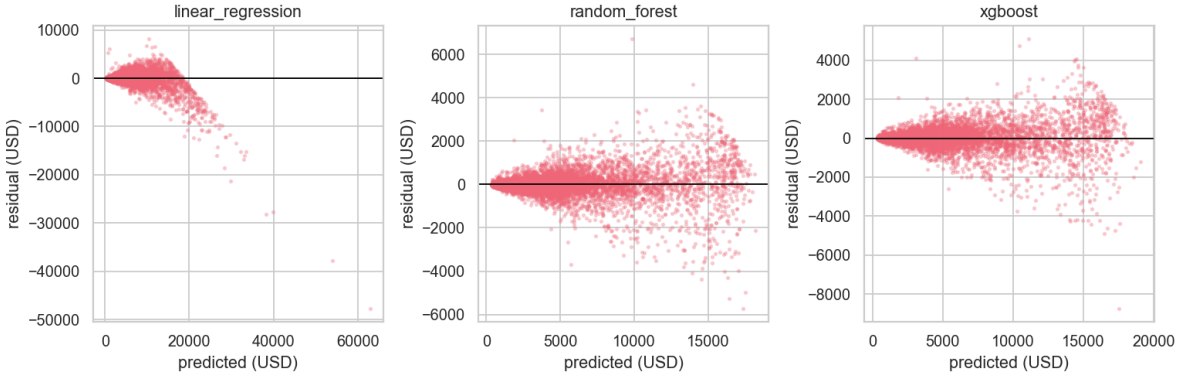


Figure 10: Cross-validated residuals for fitted models excluding the mean baseline.

The predicted-versus-actual plots confirm the ranking from the metric table. Tree ensemble

predictions stay closer to the diagonal over most of the price range, while Linear Regression shows wider deviations. The residual plots also show larger absolute errors for more expensive diamonds. This is expected for a monetary target with a wide range: even after log-target fitting, high-value diamonds remain harder to predict precisely in dollar terms.

5.1 Preliminary Test Set Evaluation

Although this is a midterm report prior to hyperparameter fine-tuning, we conducted a preliminary evaluation of our default models on the 20% held-out test split. The goal is to establish a verified baseline and ensure our cross-validation strategy is not suffering from data leakage before committing computational resources to the tuning phase.

Table 2: Preliminary evaluation results on the 20% held-out test split (default hyperparameters). Lower RMSE, MAE, and MAPE are better; higher R^2 is better.

Model	RMSE (USD)	MAE (USD)	R^2	MAPE
Mean baseline	4042.0	3057.6	-0.0001	187.7%
Linear Regression	1420.8	567.8	0.8764	12.7%
Random Forest	545.2	270.7	0.9818	6.3%
XGBoost	**537.3**	**267.9**	**0.9823**	**6.2%**

These preliminary test results closely mirror our cross-validation estimates (e.g., XGBoost RMSE of USD 537.3 vs USD 526.9). This confirms our core belief: the tree ensembles successfully capture the nonlinear dynamics of the diamond market without memorizing the training data. We can now proceed to the fine-tuning phase with confidence that our baseline is highly robust.

6 Feature Importance Discussion

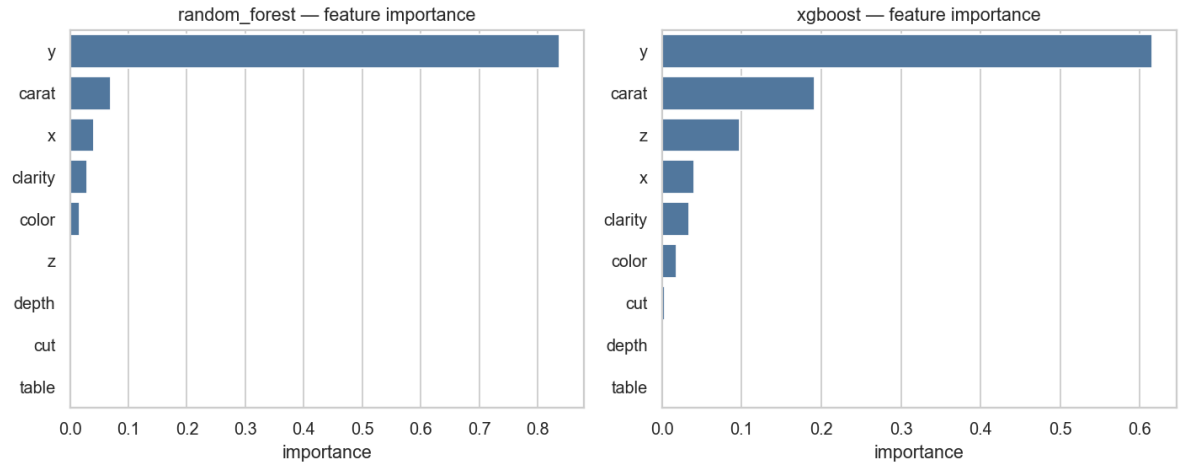


Figure 11: Model-native feature importance for Random Forest and XGBoost.

The model-native importance values are useful for diagnosis but not for causal interpretation. Random Forest assigns 83.7% of its importance to `y`, while XGBoost assigns 61.5% to `y`. `carat` is second, with 7.0% importance in Random Forest and 19.1% in XGBoost. This does not mean

that diamond width is the main economic driver of price. It reflects the collinearity seen in the EDA: `carat`, `x`, `y`, and `z` all encode stone size, and tree models can concentrate shared size information into one proxy variable.

The practical interpretation is that size dominates price prediction, but the exact feature receiving the highest model-native importance is unstable because several size variables overlap. The strongest final interpretation will focus on grouped predictors such as size, proportions, and quality grades instead of overclaiming individual size-feature effects.

7 Challenges, Limitations, and Final-Phase Implications

Developing the modeling pipeline to this midterm stage presented several analytical and engineering challenges. Chief among these was preventing data leakage, which required encapsulating all preprocessing steps strictly within Scikit-Learn Pipeline objects. Additionally, managing the strongly right-skewed target variable demanded robust transformations (via a TransformedTargetRegressor) to stabilize the training process without losing the original USD interpretability. Furthermore, because these are midterm results, The current metrics rely on fixed default hyperparameters, making the current model rankings provisional.

Looking ahead to the final phase, three primary areas of optimization have been identified to finalize the pipeline. First, given the narrow performance gap between XGBoost and Random Forest, both ensembles require systematic hyperparameter tuning, such as Grid Search or Bayesian Optimization, to ensure a fair and definitive model selection. During this midterm evaluation, diagnostic learning curves indicate that XGBoost will require stricter control over its regularization parameters, specifically the learning rate, maximum tree depth, and the total number of boosting rounds, to maximize predictive power while actively preventing late-stage overfitting. Finally, to mitigate the strong collinearity among physical size variables (`carat`, `x`, `y`, `z`), which currently fragments native feature importance and makes single-feature economic explanations unreliable, the final phase will compare two targeted strategies: consolidating the geometric dimensions into a single computed `volume` variable, versus dropping them entirely. By evaluating these configurations, we will determine which approach yields the optimal balance between predictive accuracy and model simplicity.

8 Conclusion

The midterm work establishes a reproducible prediction pipeline and produces a clear model ranking from cross-validation. The strongest result is XGBoost with RMSE USD 526.9 $\pm$ 10.3, MAE USD 262.1, $R^2 = 0.9824$, and MAPE 6.1%. The evidence supports using tree-based nonlinear models for this dataset, with XGBoost as the current selected checkpoint and Random Forest as a close alternative for the final tuning phase.